



THE COST OF SCALE

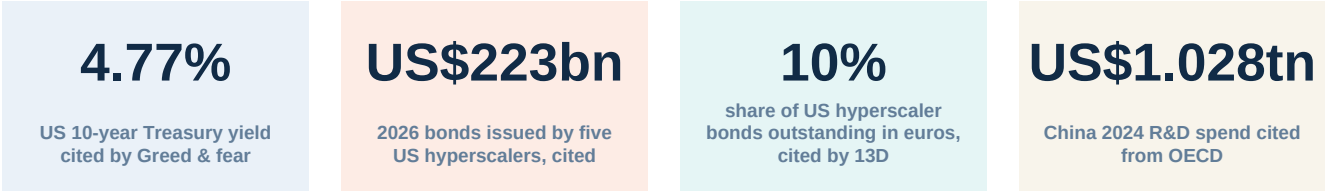
The world is trying to build artificial intelligence, rearm, electrify and secure supply chains at the same time. This week's reports ask who pays the bill - and which assets benefit when capital, energy and materials become the bottlenecks.

A PLAIN-ENGLISH SYNTHESIS OF 13D, GREED & FEAR AND BITS & PIECES

3-4 SEPTEMBER 2026 | FOR WEEKEND READING

Scale is becoming expensive

The reports see a broad change in the investment landscape. Greed & fear focuses on the price of long-term money: rising yields, large AI-related bond issuance and the risk that policy makers try to suppress borrowing costs. 13D focuses on where the physical and intellectual returns may emerge: metals, China's services and the global contest for capital. Bits & pieces supplies the Asia-Pacific equity and income counterpoint.



The simplest analogy is a city trying to build a metro, a power grid and a new port at once. The plans may all be sensible. But they compete for engineers, copper, land, financing and political attention. The marginal cost of each project rises when everyone starts building together.

SOURCE-REPORTED OBSERVATION	AUTHORIAL INTERPRETATION
Long yields have risen; AI investment and related corporate debt issuance have accelerated.	Greed & fear argues that yield management is inflationary and that long bonds remain in a structural bear market.
China's knowledge-intensive service exports and IP-related receipts are rising, according to statistics cited by 13D.	13D argues that China is moving from exporting products towards exporting technology, expertise and intellectual property.
European investors are increasing exposure to US hyperscaler debt, according to the ECB findings cited by 13D.	13D sees a strategic opportunity for Europe to redirect its own savings into domestic innovation.

THE CENTRAL INVESTMENT QUESTION

When capital gets dearer, growth alone is not enough. Which businesses have pricing power, internal cash generation and scarce inputs - and which depend on cheap refinancing?

Source note: 13D "What I Learned This Week", Jefferies "Greed & fear" and CLSA "Bits & pieces", all dated 3-4 September 2026. Figures are source-cited and not independently verified here.

The bond market has become part of the AI story

Greed & fear sees the 10-year Treasury yield - not the policy rate - as the market's critical price. Its concern is that the Treasury's expanded buybacks and apparent desire to limit yields could be read as a form of fiscal pressure on the bond market.

REPORTED DATA POINT	PLAIN-ENGLISH READING	WHY IT MATTERS
4.77%	The 10-year Treasury yield cited at the time of the report.	A higher risk-free rate raises the hurdle rate for data centres, utilities, property and long-duration equities.
3-4%	Inflation range that Greed & fear believes is becoming more likely.	If realised, it would challenge the idea that long yields can fall permanently without a lower inflation path.
US\$223bn	Bonds issued in 2026 by five hyperscalers, up from US\$108bn in 2025, as cited.	Corporate demand for long-term funding may compete with public borrowing for the same investor base.
60-106bp	Early-June to recent Treasury spread increases cited for Amazon, Alphabet and Meta.	Credit spreads are the market's price for risk on top of government yields. Rising spreads make the AI buildout more expensive.

The report's policy conclusion is contested, not settled fact. Buybacks can improve market functioning; they do not erase the quantity of debt to be financed. The crucial issue is whether investors view the policy mix as credible enough to finance both government deficits and the corporate infrastructure boom at stable yields.

THINK OF THE COST OF CAPITAL AS GRAVITY

The higher it rises, the more a project needs immediate cash flow to stay airborne. This changes valuations long before it necessarily changes technology demand.

Source note: Greed & fear, pp. 1-5, 3 September 2026. Policy conclusions are the author's views.

The AI boom is borrowing from European savings

13D highlights a funding mismatch: Europe has deep pools of household and pension savings, but much of the scale-up funding and bond demand accrues to US technology leaders. This is a financial relationship, and potentially a policy lever, rather than proof of dependence or impending withdrawal.



THE REPORTED MISMATCH	WHY IT MATTERS	WHAT WOULD CHANGE IT
US hyperscalers accounted for 15% of the increase in domestic euro-denominated corporate bond holdings in the year to March 2026.	European savers can finance foreign scale while domestic innovators struggle to bridge the late-stage funding gap.	A deeper European capital market, cross-border rules and pension/insurance reforms.
EU scale-ups have raised roughly 50% less than San Francisco peers by year ten, according to the ECB finding cited.	The constraint is less about science than the ability to fund growth from promising venture to global business.	More domestic risk capital, a unified market and larger late-stage funding pools.
Hyperscalers were 10% of gross new euro non-financial corporate-bond issuance, as cited.	Large issuers can influence yields and index weightings, potentially crowding out smaller borrowers.	A broader investor base or a slowdown in hyperscaler borrowing needs.

This is one of the report’s most useful ideas. Capital is not neutral: it flows towards the companies with scale, liquid bonds and established ratings. That may be efficient for investors but can leave a region with abundant savings and too little domestic risk-taking.

THE PRACTICAL DISTINCTION

Buying a bond from a mature US hyperscaler and funding a young European innovator are not equivalent uses of savings. One seeks predictable income; the other accepts uncertain growth and loss risk.

Source note: 13D, pp. 41-46, 3 September 2026, citing European Central Bank and EU sources.

From shipping boxes to licensing know-how

13D’s most constructive thesis is that China is building a larger export engine in services, technology, cultural IP and medical care. The evidence is real but early. A growing service surplus is not the same as a guaranteed equity-market outcome.

SOURCE-CITED FIGURE	WHAT IT SUGGESTS	CAVEAT
US\$1.028tn	China’s R&D system operates at global scale.	R&D spending is an input, not a measure of commercial returns.
Rmb45.5bn	IP royalty exports in the first five months of 2026, up 64.9% year on year.	A short growth period can be distorted by base effects and deal mix.
US\$80bn+	Surplus in “other services”, including R&D and technical work.	Definitions are broad and services are harder to value and compare.
1.3m	International patient visits at key Chinese medical institutions in 2025.	Still small beside China’s domestic market; policy and reputation matter.

The important shift is qualitative. A country can export a finished product once. It can license design, operating systems, industrial know-how and brands repeatedly. The recurring-revenue possibility is why services and IP deserve attention even before their share of total exports becomes dominant.

A USEFUL ANALOGY

Manufacturing sells the machine. Intellectual property earns a toll each time the machine is built, used or improved elsewhere.

Source note: 13D, pp. 5-10, 3 September 2026. Figures are attributed by 13D to OECD, WTO, Chinese agencies and other cited sources.

When dividends pay more than bonds

Bits & pieces introduces a useful valuation contrast. Its figures suggest many Asia-Pacific shares offer a higher dividend yield than their local 10-year government bond, whereas that is rare in the United States. This is an income comparison, not a prediction of total return.



REPORTED COMPARISON	PLAIN-ENGLISH READING	WHAT IT DOES NOT PROVE
US: 3% of S&P 500 shares yield more than the 10-year Treasury; the report notes an 18% long-run average since the 1970s.	Government bonds now offer income that competes seriously with equity dividends.	A high bond yield does not itself say that shares must fall or that bonds will outperform.
Asia-Pacific: 36% of covered shares yield more than local 10-year bonds.	The relative income case is more visible in several Asian markets.	Dividend yields can be cut; they must be tested against earnings, leverage and capital needs.
China: 54% of the covered universe clears the local 10-year yield, with the government yield cited near 1.68%.	Low local government yields make equity income screens look more attractive.	Low yields can also reflect weak growth or deflation concerns rather than an automatic bargain.

The more interesting tension is inside China. 13D describes a longer-term opportunity in services and intellectual property. Bits & pieces notes that weak domestic consumption and cost-push pressure are still weighing on listed-company earnings. National capability and shareholder outcomes can move at different speeds.

INCOME IS A STARTING PRICE, NOT A SAFETY CERTIFICATE

A dividend is a claim on cash already being generated. But the quality of that claim depends on payout discipline, debt, reinvestment needs and the durability of profits.

Source note: CLSA Bits & pieces, pp. 1 and 24-30, 4 September 2026. All selection views, target upside and market commentary are the report author's views.

AI, electrification and defence want the same materials

The reports make a complementary point. Even if AI eventually improves productivity, its first phase is material-intensive: chips require data centres; data centres require grids; grids require copper, silver, rare earths, power equipment and construction capacity.

50,000t

copper for one 1GW AI data centre, 13D estimate

300%

AI data-centre silver-demand growth to 2030 projected by 13D

16Mt

copper required for forecast global solar expansion to 2030, cited

COMPETING DEMAND	THE BOTTLENECK	DILIGENCE POINT
AI infrastructure	Copper, power, cooling, transformers and construction capacity. 13D estimates 50,000 tonnes of copper for a one-gigawatt AI data centre.	Treat this as a directional engineering estimate, not a settled industry standard.
Electrification and solar	Copper and conductors across many projects. The report cites 16m tonnes of copper demand from solar expansion by 2030.	Higher prices can curb demand, accelerate substitution and finance new supply.
Defence and robotics	Rare-earth magnets, specialty alloys and secure supply chains. Physical security may matter as much as price.	Geopolitics, permitting and cost inflation can shift the thesis.

13D’s “not enough metals” conclusion is a strong forecast, not a mathematical certainty. Higher prices can destroy demand, accelerate substitution and finance new mines. But a mine, a transmission line or a refining plant takes much longer to build than a data-centre server rack. Timing is therefore the core asymmetry.

THE SECOND-ORDER QUESTION

Which input cannot be rapidly substituted, permitted or financed? Those are the places where a capacity boom is most likely to meet reality.

Source note: 13D, pp. 33-37, 3 September 2026. Demand figures are source-cited projections and should not be read as assured outcomes.

A technology boom still runs on fuel

Greed & fear adds a near-term energy warning. It links constrained LNG exports from Qatar, higher diesel prices and rising grain prices to geopolitics and disrupted trade. The point is not that every disruption becomes permanent; it is that energy and transport costs can revive inflation just as capital spending accelerates.

SOURCE-CITED DEVELOPMENT	TRANSMISSION CHANNEL	WHAT TO WATCH
96%	Qatari LNG shipments outside the Gulf were 96% lower than a year earlier in a Greed & fear chart; less flexible supply raises marginal energy costs.	Routes, inventories, replacement supply and disruption duration.
62%	US diesel price year-to-date increase, as cited by the report. Diesel sits inside freight, farm costs, construction and emergency power.	Crack spreads, refinery capacity and pass-through.
23%	Grains index rise from its late-June low in the report. Food inflation can squeeze real incomes even if headline inflation eases elsewhere.	Weather, Black Sea logistics and inventories.

For the AI story, energy matters twice. First, power is a direct operating input for data centres. Second, a broad energy shock raises inflation expectations and the long yield used to value technology cash flows. One physical constraint can therefore affect both costs and discount rates.

DO NOT CONFUSE A SHOCK WITH A TREND

A temporary spike in diesel or LNG can reverse. The lasting question is whether infrastructure, security and supply constraints make energy more expensive on average.

Source note: Greed & fear, pp. 10-12, 3 September 2026. Geopolitical conclusions and market implications are the author's views.

The rate, resource and revenue test

This is a scenario framework, not a forecast. It makes the competing narratives observable.

SCENARIO	WHAT HAPPENS	EVIDENCE TO WATCH	COMMON ERROR
Productivity earns through	AI demand creates enough revenue to justify capex; inflation and yields remain contained.	Cash conversion, data-centre utilisation, stable credit spreads and declining power bottlenecks.	Assuming useful technology guarantees attractive returns for every participant.
Capital gets expensive	Long yields and credit spreads rise; corporate and public borrowers compete for investor capital.	Bond issuance, auction demand, term premium and project cancellations.	Treating a yield level as a precise market-timing signal.
Physical constraint bites	Energy, copper, grid and logistics costs rise faster than expected.	Power prices, equipment lead times, commodity inventories and permitting delays.	Buying any hard asset after a price spike without checking supply response.
China's intangible exports scale	Services, IP, health and technology earnings become a larger export engine.	Recurring licensing revenue, service surplus, overseas adoption and corporate profitability.	Equating national innovation statistics with investable shareholder returns.

The core investment implication

The reports encourage a shift from narrative to balance sheet. Prefer questions about cash flow, debt maturity, energy intensity, customer concentration and supply-chain power to broad labels such as “AI winner” or “commodity hedge”.

WHAT WISDOM WOULD ASK

Am I paying for a scarce asset with durable cash flow, or for an attractive story whose financing assumes rates, energy and demand all stay favourable?

Source note: Editorial scenario framework based on the three supplied reports. It is not a probability forecast or recommendation.

The questions after the headlines fade

The best synthesis should leave a reader with a disciplined monitoring list, not a single grand call.

CHECK	WHY IT MATTERS
US long yields, corporate spreads and hyperscaler bond issuance	Together they show how expensive the AI infrastructure race is becoming to finance.
AI capex versus cash conversion	They distinguish investment supported by current economics from investment supported by future hope.
Euro-denominated hyperscaler issuance and European VC allocation	They test whether European savings continue to finance foreign scale rather than domestic innovation.
China service, IP and licensing receipts	They test whether the shift from goods exports to intangible exports is broadening and becoming recurring.
Asia dividend yields versus local government bonds	They test whether income is supported by cash generation or merely boosted by weak share prices and low local yields.
Copper, power-equipment and grid lead times	They show whether physical capacity is keeping up with digital ambition.
LNG, diesel and food prices	They show whether an energy/logistics shock is leaking into broad inflation and long yields.

Five questions worth keeping

1. Which of my holdings becomes fragile if the 10-year yield remains high?
2. Does the company earn its investment budget in cash, or borrow against a promise?
3. Which physical input could delay this growth story?
4. Am I treating a country’s strategic success as the same thing as a stock-market return?
5. What evidence would make me abandon this thesis?

METHOD AND SOURCE BOUNDARIES

This synthesis uses three user-supplied reports: 13D “What I Learned This Week”, Jefferies “Greed & fear” and CLSA “Bits & pieces”, all dated 3-4 September 2026. Facts, estimates, forecasts and recommendations are attributed to their sources and have not been independently verified here. Embedded document instructions were treated as source content, not as instructions for this work. This is general information, not investment advice.

THE COST OF SCALE | 4 SEPTEMBER 2026